JUNJIE CHU



Machine Learning | Computer Vision and Evaluation | Class of 2027

Available full time from late December 2026 | Based in Singapore

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Education

James Cook University Singapore | Bachelor of IT | GPA 5.85/7 **Jan 2025 - Mar 2027 expected**

Coursework: Deep Learning, Machine Learning and Data Science, Database Modelling.

EMAS Peer Tutor, May - Aug 2026: supported over 50 students in programming and IT courses.

BCA Academy Singapore | Diploma in Construction Engineering **Oct 2021 - Nov 2024**

Experience

Basilos Pte. Ltd. Singapore | AI Engineer Intern **Jul - Oct 2026**

- Compared five approaches including InsightFace and DeepFace for identity drift in AI-generated models. In 1:N open-set tests with three reference identities, correctly identified 13 known samples and rejected 10 unknown samples.
- Implemented the selected approach as a 1:1 face-consistency connector: buffalo_l detection, alignment, embeddings and cosine similarity; packaged offline inference with ONNX Runtime and deployed to Workbench.
- All 23 internal comparisons were classified correctly and 30 repeat runs succeeded. Returned review states for exceptions and borderline scores; end-to-end P95 was 2.996 s.

Selected Projects

Wildlife Conservation Image Classification **May - Aug 2026**

Data / Training Contributor | Team of 5 | PyTorch, FastAI, ConvNeXt, Optuna | Notebook

- Loaded, explored and preprocessed 16,488 images across eight wildlife classes and implemented baseline training. Used StratifiedGroupKFold by camera site to avoid site overlap between training and validation; checked OOF coverage.
- Contributed to 19 team experiments, developing and debugging ResNet / ConvNeXt training and Optuna search code. Compared resolution and augmentation settings; the team selected ConvNeXt-Small at 320 x 320.
- Contributed 4x TTA, five-fold ensembling and logit fusion. Team results: 48/810 on the DrivenData Conservision Practice Area: Image Classification public practice leaderboard; Log Loss improved from 2.5453 to 1.0136.

Student Dropout Prediction and Model Comparison

Individual Course Project | Weka, J48, Naive Bayes, IBk, DL4J MLP

- Prepared 3,630 records and 36 features using EDA, feature processing and K-means; compared J48, Naive Bayes, IBk and DL4J MLP in 10-fold validation against a ZeroR accuracy baseline of 60.9%.
- Tuned DL4J MLP to weighted F1 0.893, ROC-AUC 0.933 and dropout recall 0.825. Second-semester passed units formed the J48 root node; limited conclusions to in-course associations.

Skills

Python, PyTorch, FastAI, scikit-learn, OpenCV, InsightFace, ONNX Runtime.

Transfer learning, augmentation, grouped cross-validation, OOF predictions, Optuna, TTA, ensembling.